Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/18/2018
PI Name:	Sadek, Hesham M.D., Ph.D.		
Project Title:	Long-Term Effect of Space Radiation on Cardiomyocyte Turnover		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hesham.sadek@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214-648-1413
Organization Name:	University of Texas Southwestern Medical Center at Dallas		
PI Address 1:	Internal Medicine/Cardiology, NB10.222		
PI Address 2:	6000 Harry Hines Blvd.		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75235-5303	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/20/2015	**End Date:**	01/19/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/19/2018 per S. Monk/LaRC (Ed.,1/13/17)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Asaithamby, Aroumougame Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX15AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The risks to astronauts from exposure to space radiation have long been of great concern to NASA. These risks are now more important with the advent of long-duration space missions, which raise the risk of exposure to space radiation. While the effect of space radiation on cancer and coronary artery disease has been studied, little is known about the effect of space radiation on the heart muscle, in particular on the ability of the heart muscle to replenish itself. The adult heart is incapable of regenerating itself after injury because the vast majority of heart muscle cells cannot divide. Nevertheless, the adult human heart slowly replenishes 1-2% of its muscle every year, and it is estimated that in a human heart close to 50% of the heart muscle is renewed during its lifespan. Therefore, preventing the formation of new heart muscle cells could result in a loss of up to 50% of heart muscle leading to heart failure. We recently discovered that after birth, the increase in oxygen in the atmosphere results in formation of harmful reactive oxygen species (ROS), which in turn damage the DNA of heart muscle cells (Cardiomyocytes) and cause them to stop dividing. Intriguingly, a small population of dividing heart muscle cells is protected from DNA damage by maintaining themselves in a low oxygen environment within the heart. However, if prolonged exposure to space radiation damages the DNA of these cells, then the heart will lose any ability to replenish itself. Our proposal will determine the long-term effect of space radiation on the heart by shutting down the slow, but crucial, ability of the heart to replenish itself. The long-term goal of this project is to develop countermeasures to reduce heart failure risk due to space radiation in Astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research would also lead to the development of countermeasures to enhance astronaut's ability to sustain HZE (high energy ions) particles exposures in space.
Task Progress:	1. Iron particles-induced complex double-strand breaks (DSBs) are irreparable in cardiomyocytes. 2. Iron particles trigger cardiac hypertrophy in mice. 3. Both fractionated (20 cGy X 5) and acute (100 cGy) doses of 600 MeV/n iron particles significantly increase the ejection fraction of mouse heart ~1 year after the iron particle irradiation.
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Abstracts for Journals and Proceedings	Asaithamby A, Sadek HA. "Space Radiation Triggers Cardiac Hypertrophy." 28th Annual Space Radiation Investigators' Workshop, Galveston, TX, January 23-26, 2017. 28th Annual Space Radiation Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Articles in Peer-reviewed Journals	Nakada Y, Nhi Nguyen NU, Xiao F, Savla JJ, Lam NT, Abdisalaam S, Bhattacharya S, Mukherjee S, Asaithamby A, Gillette TG, Hill JA, Sadek HA. "DNA damage response mediates pressure overload-induced cardiomyocyte hypertrophy. " Circulation. 2019 Feb 26;139(9):1237-9. https://doi.org/10.1161/CIRCULATIONAHA.118.034822 ; PubMed PMID: 30802166 , Feb-2019
Articles in Peer-reviewed Journals	Cardoso AC, Lam NT, Savla JJ, Nakada Y, Pereira AHM, Elnwasany A, Menendez-Montes I, Ensley EL, Petric UB, Sharma G, Sherry AD, Malloy CR, Khemtong C, Kinter MT, Tan WLW, Anene-Nzelu CG, Foo RS, Nguyen NUN, Li S, Ahmed MS, Elhelaly WM, Abdisalaam S, Asaithamby A, Xing C, Kanchwala M, Vale G, Eckert KM, Mitsche MA, McDonald JG, Hill JA, Huang L, Shaul PW, Szweda LI, Sadek HA. "Mitochondrial substrate utilization regulates cardiomyocyte cell cycle progression." Nat Metab. 2020 Feb;2(2):167-78. https://doi.org/10.1038/s42255-020-0169-x ; PMID: 32617517; PMCID: PMC7331943 , Feb-2020
Articles in Peer-reviewed Journals	Sadek HA, Porrello ER. "Neonatal heart regeneration: Moving from phenomenology to regenerative medicine." J Thorac Cardiovasc Surg. 2020 Jun;159(6):2451-5. https://doi.org/10.1016/j.jtcvs.2019.11.140 ; PMID: 32087960 , Jun-2020
Articles in Peer-reviewed Journals	Bhattacharya S, Srinivasan K, Abdisalaam S, Su F, Raj P, Dozmorov I, Mishra R, Wakeland EK, Ghose S, Mukherjee S, Asaithamby A. "RAD51 interconnects between DNA replication, DNA repair and immunity." Nucleic Acids Res. 2017 May 5;45(8):4590-605. https://doi.org/10.1093/nar/gkx126 ; PubMed PMID: 28334891; PubMed Central PMCID: PMC5416901 , May-2017
Articles in Peer-reviewed Journals	Bhattacharya S, Asaithamby A. "Repurposing DNA repair factors to eradicate tumor cells upon radiotherapy." Transl Can Res. 2017 Jul;6 Suppl 5:S822-S839. Review. https://doi.org/10.21037/tcr.2017.05.22 , Jul-2017